

Location Estimation Using Differential RSS with Spatially Correlated Shadowing

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Abstract—In this paper, we propose a new localization technique using differential received signal strength (DRSS) which does not require signal source cooperation for location estimation. Specifically, we introduce a DRSS-based localization framework as well as its geometric interpretation for both local and global positioning to facilitate understanding of the approach. Then, a least-squares (LS) optimization framework is formulated for DRSS-based location estimation (DRLE) which makes full use of the DRSS measurements available. Our study shows that the localization performance of DRLE and its RSS-based counterpart (RLE) is substantially affected by spatially correlated shadow fading under which their localization behaviors are found to be different. Finally, we argue that DRLE has practical advantages over other positioning techniques, and show that its location accuracy is comparable or even superior to RLE.

I. INTRODUCTION

Location estimation has become an important task due to the popularity of wireless sensor networks and increasing demand for location-based services. In range-based techniques, the range estimates used for source localization are typically based on either received signal strength (RSS), time-of-arrival (TOA), time-difference-of-arrival (TDOA), angle-of-arrival (AOA), or their combinations [1], [2].

Despite lower location accuracy with a small number of nodes in general, RSS-based localization is a low-complexity, cost-effective solution. Specifically, it is attractive because RSS is readily available in most types of wireless systems, and may be the only ranging information available in some scenarios, for example, in severe indoor/urban multipath environments or in surveillance applications. However, RSS-based approaches require accurate knowledge of transmitter and environmental parameters such as the transmitted power and system/propagation loss factors [3]. The uncertainty existing at the receiver end substantially increases its reliance on labor-intensive offline calibration of the parameters and explicit notifications from a source via a predefined beacon/pilot or handshake-based protocol. Further, practical implementation issues cause some degree of discrepancy between the actual calibrated parameter values and their estimates [4].

In this paper, we propose a least-squares (LS) localization technique based on differential RSS (DRSS), termed DRSS-based location estimation (DRLE), which makes full use of the redundant DRSS information available. DRSS is defined as the difference in logarithmic received power levels (*i.e.*, ratio in linear) of a transmitting target (source) at different

anchor locations. Due to the use of DRSS, the requirements for accurate parameter information can be removed or significantly relaxed. Some related work can be found in the context of cellular networks [5], [6]. We compare DRLE with its RSS-based localization counterpart (RLE) under realistic correlated shadowing conditions, where the localization behavior of the two methods is shown to be very different. For ease of demonstration, the two-dimensional single source localization problem is presented. However, the approach can readily be extended to three dimensions and to multiple sources.

The paper is organized as follows. Section II is devoted to a presentation of the assumed radio link environment and the observables used for DRLE. In Section III, we introduce the DRSS-based localization framework and present a geometric interpretation of the approach. Section IV presents the details of an LS optimization framework for DRLE. Next, we provide comparisons between DRLE and RLE in Section V. Finally, Section VI concludes the paper.

II. LOG-DISTANCE PATH LOSS MODEL WITH CORRELATED SHADOWING

In RSS-based source localization systems, m reference or anchor nodes with known coordinates estimate their distances to the source based on the measured RSS. The distance estimation is generally based on a specific large-scale radio propagation model. As many measurement campaigns [3], [7] and analytical results [8] have shown, the RSS-distance relationship along with the environment-dependent uncertainties can be captured in a general log-distance relationship. Specifically, the observed loss in signal strength at the i th anchor can be written as

$$v_i \triangleq P_r(d_0) \text{ (dBm)} - P_r(d_i) \text{ (dBm)} \\ = L(d_i; \theta) + X_{\sigma_i}, \quad i = 1, \dots, m, \quad (1)$$

where $P_r(d_i)$ is the RSS at the i th anchor, $P_r(d_0)$ is power received at a close-in reference distance d_0 , $L(d_i; \theta) = 10\alpha(\log_{10} d_i - \log_{10} d_0)$ denotes the log-distance path loss. Here α is the path loss gradient. The distance between the source and i th anchor node is denoted by d_i , while θ is a vector of unknown parameters to be estimated including the source's coordinates and radio dependent parameters. X_{σ} is empirically modeled as a Gaussian random variable with zero mean and variance σ_S^2 corresponding to large-scale shadow fading.

This environment-dependent variability is one of the most influential yet unavoidable factors in RSS-based localization.

The received signal power $P_r(d_0)$ in Eq. (1), predicted by an empirical model or measured at the reference distance d_0 , depends primarily on two types of parameters: transceiver and environmental parameters [3]. To minimize localization error, it is essential to know or estimate these parameters as precisely as possible, thus requiring an offline calibration effort based on measurements. However, in many practical situations, this manual effort may be too costly or infeasible. Even if the environmental parameters can accurately be determined and known to anchor nodes, the transmitter parameter values may not be readily available for the anchors, or could be erroneously reported or even disguised. In most proposed RSS-based positioning techniques, the values are assumed to be perfectly known *a priori*.

One means of eliminating or reducing the need for knowledge of the transmitter and environmental parameters is to change the observation to *differential* RSS:

$$\begin{aligned} v_{ij} &\triangleq v_j - v_i \\ &= P(d_i, d_j) - K_{ij} \\ &= L(d_i, d_j; \boldsymbol{\theta}) + \Delta X_{\sigma_{ij}}, \end{aligned} \quad (2)$$

where the DRSS measurement $P(d_i, d_j)$ and differential log-distance path loss $L(d_i, d_j; \boldsymbol{\theta})$ (hereafter, $L_{ij}(\boldsymbol{\theta})$) are defined by

$$P(d_i, d_j) \triangleq P_r(d_i) - P_r(d_j) \quad (3)$$

and

$$L(d_i, d_j; \boldsymbol{\theta}) \triangleq 10\alpha(\log_{10} d_j - \log_{10} d_i), \quad (4)$$

respectively. Here $i, j \in \{1, \dots, m\}$, $i < j$. As a general rule of notation in this paper, the subscripts fall in this range, unless indicated otherwise. $\boldsymbol{\theta}$ is the source coordinates $\mathbf{x}_s = [x_s, y_s]^T$, and K_{ij} is a deterministic variable including the receiver-specific parameters which can be accurately determined. It can be noticed that the transmitter uncertainties in $P_r(d_0)$ are removed. Also, note that RSS measurements at m anchor nodes yield an un-ordered set of

$$M = \binom{m}{2} = \frac{m(m-1)}{2} \quad (5)$$

distinct DRSS measurements and corresponding differential path loss equations, where an ij -pair and ji -pair are counted only once. This set is divided into $m-1$ *basic* and $\frac{(m-1)(m-2)}{2}$ *redundant* measurements, meaning that the whole set of size M can be determined by a linear combination of the $m-1$ basic measurements. The geometric interpretation of these equations for localization is presented in Section III-B.

The difference between the two shadowing random variables $X_{\sigma_j} - X_{\sigma_i}$ is denoted by $\Delta X_{\sigma_{ij}}$. It is clear from Eq. (1) that, because of the log-normal shadowing X_{σ} , the received signal power at the i th reference location $P_r(d_i)$ is also a Gaussian random variable (conditioned on the distance d_i). Accordingly, in Eq. (2), the random variables $P_r(d_i)$ and $P_r(d_j)$ from the same source are jointly Gaussian. Specifically, the two random variables representing shadowing X_{σ_i} and X_{σ_j} have a joint

probability density function (PDF)

$$\begin{aligned} f_{X_{\sigma_i}, X_{\sigma_j}}(\eta_i, \eta_j) &= \frac{1}{2\pi\sigma_S^2\sqrt{1-\rho_{S_{ij}}^2}} \\ &\cdot \exp \frac{-\left(\eta_i^2 - 2\rho_{S_{ij}}\eta_i\eta_j + \eta_j^2\right)}{2\sigma_S^2(1-\rho_{S_{ij}}^2)}, \end{aligned}$$

which leads to the PDF of $\Delta X_{\sigma_{ij}}$

$$\begin{aligned} f_{\Delta X_{\sigma}}(\Delta\eta_{ij}) &= \frac{1}{2\pi\sigma_S^2\sqrt{1-\rho_{S_{ij}}^2}} \\ &\cdot \int_{-\infty}^{+\infty} \exp \frac{-\left(\eta_i^2 - 2\rho_{S_{ij}}\eta_i(\eta_i - \Delta\eta_{ij}) + (\eta_i - \Delta\eta_{ij})^2\right)}{2\sigma_S^2(1-\rho_{S_{ij}}^2)} d\eta_i \\ &= \frac{1}{2\sigma_S\sqrt{\pi(1-\rho_{S_{ij}})}} \exp \left(\frac{-\Delta\eta_{ij}^2}{4\sigma_S^2(1-\rho_{S_{ij}})} \right). \end{aligned}$$

Thus, $\Delta X_{\sigma}(\Delta\eta_{ij})$ is Gaussian with zero mean and variance $\hat{\sigma}_S^2 = 2(1-\rho_{S_{ij}})\sigma_S^2$. $\rho_{S_{ij}}$ is the correlation coefficient reflecting the degree of spatial correlation between the shadow fading at any two locations i, j since $\hat{\sigma}_S^2 = 2\sigma_S^2 - 2C(X_{\sigma_i}, X_{\sigma_j})$ where the covariance $C(X_{\sigma_i}, X_{\sigma_j}) = \rho_{S_{ij}}\sigma_S^2$. The observation v_{ij} in Eq. (2) is thus Gaussian with PDF

$$f_V(v_{ij}; \boldsymbol{\theta}) = \frac{1}{2\sigma_S\sqrt{\pi(1-\rho_{S_{ij}})}} \exp \left(\frac{-(v_{ij} - L_{ij}(\boldsymbol{\theta}))^2}{4\sigma_S^2(1-\rho_{S_{ij}})} \right).$$

Although most existing studies in RSS-based localization simply assume that shadowing noise components at two locations are independent (*i.e.*, $\rho_S = 0$), in reality the spatial correlation of shadow fading is often substantial due to similar terrain configuration or obstacles on signal propagation paths between the source and anchor nodes. It has been found in previous empirical studies that a typical value of the correlation coefficient ranges from 0.2 to 0.8 in indoor [7] and outdoor [9] channels, and as the angle and distance between a pair of reference locations decreases, the correlation tends to increase. Therefore, it is important to take the correlation into account for the analysis of RSS-based location estimation.

In this paper, we generate a correlated shadowing vector $\mathbf{X}_{\sigma} = [X_{\sigma_1}, \dots, X_{\sigma_m}]^T$ in Eq. (1) with an $m \times m$ covariance matrix $\mathbf{K}$ whose ij element (assuming $\rho_{S_{ij}} = \rho_{S_{ji}}$)

$$K_{ij} = \begin{cases} \sigma_S^2 & \text{if } i = j \\ \rho_{S_{ij}}\sigma_S^2 & \text{if } i \neq j, \end{cases}$$

using a vector $\mathbf{w} = [w_1, \dots, w_m]^T$ of m zero-mean, unit-variance, uncorrelated random variables. The symmetric and nonnegative definite matrix $\mathbf{K}$ is then decomposed into $\mathbf{K} = \mathbf{L}\mathbf{L}^T$ by means of Cholesky factorization to obtain $\mathbf{X}_{\sigma} = \mathbf{L}\mathbf{w}$ [10]. It should be noted that as the correlation increases, the overall shadowing variance of $\Delta X_{\sigma_{ij}}$ in Eq. (2) decreases.

III. GEOMETRIC INTERPRETATION OF DRSS-BASED POSITIONING

In this section, we examine DRSS-based positioning from a geometric perspective. Specifically, new geometric formulations for local and global positioning are presented. Interestingly, as multilateration systems using TDOA create

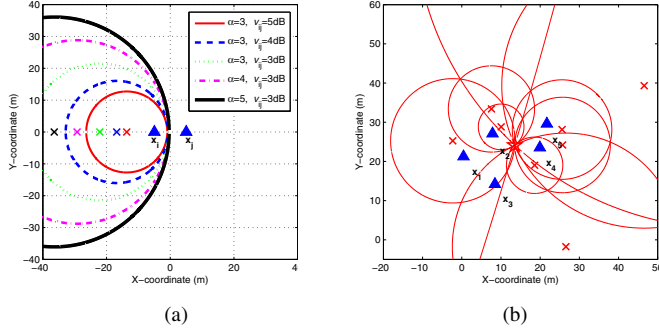


Fig. 1: (a) Local and (b) global geometric interpretation of DRSS-based positioning (noiseless case). The source, anchor and the center of each geometric DRSS circle are indicated by “★”, “▲” and “×”, respectively.

hyperbolic curves, DRSS also forms a geometric system of nonlinear functions, but its geometric curves are circular. In the absence of shadowing and measurement noise, it is shown that the source is located at the intersection of these circles. Despite the similarity of the geometric shapes to those found in trilateration approaches [1], we show here that the geometry and redundancy of the circles are very different.

Geometric source positioning algorithms can be comprised of four phases (esp. for WSNs): 1) Each pair of unlocalized nodes estimates their inter-node distance; 2) By measuring a source’s signal feature of interest (RSS, DRSS, TOA, etc), the locus of points on which the source resides is formed in each pair’s local coordinate system; 3) Then, multiple local coordinate systems are integrated into the single *relative* coordinate system which is a rotated, mirrored, and/or scaled version of the original absolute coordinate system; 4) When absolute coordinates of a portion of the nodes (usually ≥ 3) is known, the relative location map is converted into the global location map via a geometric transformation. The first three phases are termed local/relative positioning whereas the last task enables global positioning. In the following we discuss phases 1, 2 and 4 for DRSS-based positioning. Due to the space constraints, we do not discuss phase 3 here and refer the interested readers to the literature (e.g., [11]) for the details.

A. Geometrical Formulation for Local/Relative Positioning

Our aim here is to determine local coordinates $\mathbf{X}_s = [X_s, Y_s]^T$ of some unknown source (e.g., defective) *without* any cooperation from the source. Specifically, we describe how each pair of sensors, having their inter-node distance estimates but without knowledge of their global coordinates, creates a local coordinate system where a nonlinear geometric function is formed using DRSS measurements.

Let us first suppose that each node i has measured its distance $D_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ to neighbors j based on some reliable range estimates, where $\{\mathbf{x}_i\}_{i=1}^m$ denotes the i th *unknown* absolute coordinate vector $\mathbf{x}_i = [x_i, y_i]^T$, and $\|\cdot\|_2$ is the L_2 norm. Next, the middle of the link connecting the pair is translated to the origin of the local coordinate system (X, Y) for the pair. Thus, the two nodes are located at $(-\frac{D_{ij}}{2}, 0)$ and

$(\frac{D_{ij}}{2}, 0)$ on the local x-axis, respectively. This local coordinate system produces a geometric function of $L_{ij}(\boldsymbol{\theta})$ using Eq. (3):

$$\begin{aligned} L_{ij}(\boldsymbol{\theta}) &\triangleq 5\alpha \left\{ \log_{10} \left((X - \frac{D_{ij}}{2})^2 + Y^2 \right) \right. \\ &\quad \left. - \log_{10} \left((X + \frac{D_{ij}}{2})^2 + Y^2 \right) \right\} \\ &= 5\alpha \log_{10} \left(1 - \frac{2D_{ij}X}{(X + \frac{D_{ij}}{2})^2 + Y^2} \right). \end{aligned}$$

Then, the estimate of a local y -coordinate of the source location can be found to be on

$$Y = \pm \sqrt{\frac{2D_{ij}X}{1 - h_{ij}} - \left(X + \frac{D_{ij}}{2}\right)^2},$$

where $h_{ij} = 10^{v_{ij}/5\alpha}$. As shown in Fig. 1a, in the absence of noise the pair of nodes (plotted as triangles) form a *local circle* on which the unknown source is located, as:

$$\left(X + \frac{D_{ij}}{2} \left(\frac{h_{ij} + 1}{h_{ij} - 1}\right)\right)^2 + Y^2 = \frac{h_{ij} D_{ij}^2}{(h_{ij} - 1)^2}. \quad (6)$$

The inter-node distance D_{ij} is assumed to be 10 meters in Fig. 1a. As noted, the focus of the circle (marked as “×” in the figure) is located at $(-\frac{D_{ij}}{2} \left(\frac{h_{ij} + 1}{h_{ij} - 1}\right), 0)$ on the x-axis in the local coordinate system. Clearly, the geometry (focus and radius) of the circles is different from that in RSS/TOA-based trilateration, where the centers of the circles are located at the sensor positions. It is also noted that the circles are impacted by the path loss gradient α and observation v_{ij} as well as the relative distance D_{ij} . As shown in Fig. 1a, the circle enlarges as either v_{ij} decreases or α increases. This is because, for fixed α and decreasing v_{ij} (i.e., DRSS $P(d_i, d_j)$ in Eq. (3) is decreasing), the source is further from the i th node *relative* to the distance to the j th node. On the other hand, for fixed v_{ij} and smaller α , h_{ij} becomes larger for the same DRSS value so that the pattern is reversed. It can be noticed from Eq. (6) that only v_{ij} and the relative distance D_{ij} are needed for local positioning. Finally, it should be noted that unlike RSS/TOA-based trilateration, where each measurement results in a circle, here two sensor measurements correspond to a single circle.

B. Geometrical Formulation for Global Positioning

Although simply knowing the relative locations of the source will suffice for some applications such as WSNs/CRNs, global knowledge of the source position is essential in many scenarios. Given m absolute coordinates for the anchor locations $\{\mathbf{x}_i\}_{i=1}^m$, we next describe how to transform the local geometric systems built by pairs of neighboring nodes into a global geometric system via a linear transformation. This is considered as a linear mapping in the form of $\tilde{\mathbf{x}}_s = \mathbf{T}^{(k)} \tilde{\mathbf{X}}_s$, $k = 1, \dots, M$, where in the problem of single source location¹, we use the affine transformation

$$\tilde{\mathbf{x}}_s \triangleq \begin{pmatrix} \mathbf{x}_s \\ 1 \end{pmatrix}_{3 \times 1}, \mathbf{T}^{(k)} \triangleq \begin{pmatrix} t_{11} & t_{12} & t_{13} \\ t_{21} & t_{22} & t_{23} \\ 0 & 0 & 1 \end{pmatrix}_{3 \times 3}, \tilde{\mathbf{X}}_s \triangleq \begin{pmatrix} \mathbf{X}_s \\ 1 \end{pmatrix}_{3 \times 1}$$

¹For multiple source localization, a unique solution usually cannot be found so that an optimization scheme (e.g., least squares) needs to be employed.

where $t_{11} = \cos(\varphi_{ij})$, $t_{21} = \sin(\varphi_{ij})$, $t_{12} = -\sin(\varphi_{ij})$, and $t_{22} = \cos(\varphi_{ij})$ are the rotation elements. Let $x_{ij} = x_j - x_i$ and $y_{ij} = y_j - y_i$. Then $\varphi_{ij} = \arctan\left(\frac{y_{ij}}{x_{ij}}\right)$ is the angle of the vector pointing from the i th anchor to the j th anchor relative to the x -axis counterclockwise. The translation elements $t_{13} = \frac{1}{2}(x_i + x_j)$ and $t_{23} = \frac{1}{2}(y_i + y_j)$ denote the x and y coordinates of the center of the link connecting an ij -pair of anchors, respectively. If v_{ij} is negative, the circles are reflected.

In Fig. 1b, using different $M = \binom{5}{2} = 10$ pairs of nodes, ten *global circles* with different geometries are formed which all intersect at the source position (x_s, y_s) . Each circle has its center located at $c_{dk} = (x_{dk}, y_{dk})$ where

$$x_{dk} \triangleq \frac{h_{ij}x_i - x_j}{h_{ij} - 1}, \quad y_{dk} \triangleq \frac{h_{ij}y_i - y_j}{h_{ij} - 1}, \quad (7)$$

and has a radius r_{dk} equal to

$$r_{dk} \triangleq \frac{\sqrt{h_{ij} \cdot D_{ij}}}{|h_{ij} - 1|}, \quad k = 1, 2, \dots, M$$

where $|\cdot|$ denotes the absolute value. Unlike the local circle in Eq. (6), the center of the global circle is affected by the absolute coordinates. Note that in the presence of noise, both the circle's radius and focus will be impacted unlike in well-known range-based circular positioning methods.

Let us next look into the geometric implication of the DRSS redundancy discussed in Section II. In the simple noiseless example with three anchors ($m = 3$), we discussed earlier that two basic and one redundant measurements exist. Accordingly, using Eq. (7) we can create three distinct geometric circles, yet only the two associated with the basic measurements are independent. This is reflected by the fact that the circle associated with the redundant measurement intersects both of the other circles at the same two points at which the two independent circles meet. Thus, we need a fourth anchor in order to have an unambiguous solution. Despite the need for an additional independent measurement as compared with other circular positioning methods, in the presence of noise the redundant DRSS measurements of higher rate of increase with additional anchors increases robustness against noise as demonstrated in Section V.

IV. LS LOCATION ESTIMATION USING DRSS

In the presence of noise, the circles defined by DRSS measurements will not intersect at a common point. As a result, we have an overdetermined system of nonlinear equations, and a solution does not generally exist. In such a case, the classic approach is to find the solution which minimizes the square error between the observations and the assumed relationship described by Eq. (2), *i.e.*, the least-squares (LS) solution. This section is devoted to formulating an LS optimization problem for DRLE and discussing numerical algorithms suitable for solving the problem.

A. Formulation of Least-Squares Optimization

Consider the following LS optimization problem that estimates source coordinates $\theta = [x_s, y_s]^T$ based on observed v_{ij} ,

assuming *a priori* knowledge of the path loss gradient².

$$\hat{\theta} = \min_{\theta} \left\{ \phi(\theta) \triangleq \frac{1}{2} \sum_{i,j \in \{1, \dots, m\}} r_{ij}^2(\theta) \right\} \quad (8)$$

where the residual $r_{ij}(\theta) \triangleq v_{ij} - L_{ij}(\theta) : \mathbb{R}^2 \rightarrow \mathbb{R}$, constructing the residual vector $\mathbf{r} : \mathbb{R}^2 \rightarrow \mathbb{R}^M$ in the form of $\mathbf{r}(\theta) = [r_{12}, r_{13}, \dots, r_{1m}, r_{23}, \dots, r_{(m-1)m}]^T$. Note that our LS framework for DRLE incorporates the redundant DRSS measurements. Using this vector form, we can rewrite the smooth nonlinear objective function ϕ as $\phi(\theta) \triangleq \frac{1}{2} \|\mathbf{r}(\theta)\|_2^2$. The first derivative of ϕ with respect to θ is

$$\nabla \phi(\theta) = \sum_{i=1}^{m-1} \sum_{j=i+1}^m r_{ij}(\theta) \nabla r_{ij}(\theta) = -\mathbf{J}(\theta)^T \mathbf{r}(\theta),$$

in which $\mathbf{J}$ is the $M \times 2$ Jacobian matrix of the vector $\mathbf{L}(\theta) = [L_{12}, \dots, L_{1m}, L_{23}, \dots, L_{(m-1)m}]^T$ defined by

$$\mathbf{J}(\theta) \triangleq \left[\frac{\partial L_{ij}(\theta)}{\partial \theta_k} \right]_{\substack{i,j \in \{1, \dots, m\}, \\ k=1,2}}.$$

The 2-row vector $\mathbf{J}$ is found to be

$$\mathbf{J} = 10\alpha \log_{10} e \cdot \left[\frac{\bar{x}_j}{d_j} - \frac{\bar{x}_i}{d_i}, \frac{\bar{y}_j}{d_j} - \frac{\bar{y}_i}{d_i} \right]_{\substack{i,j \in \{1, \dots, m\}, \\ i < j}}$$

where

$$\begin{aligned} \bar{x}_i &\triangleq \frac{x - x_i}{d_i}, \quad \bar{x}_j \triangleq \frac{x - x_j}{d_j}, \\ \bar{y}_i &\triangleq \frac{y - y_i}{d_i}, \quad \bar{y}_j \triangleq \frac{y - y_j}{d_j}. \end{aligned}$$

Here d_i is the distance estimate to the source from anchor node i . Geometrically, $\bar{x}_i$ and $\bar{y}_i$ are x and y elements of the unit position vector $\mathbf{u}_i \triangleq \bar{x}_i \mathbf{e}_x + \bar{y}_i \mathbf{e}_y$ in the Cartesian coordinate system, where $\mathbf{e}_x$ and $\mathbf{e}_y$ are unit vectors in the direction of x and y axes, respectively. The position vector represents the direction from the i th anchor to the source node.

B. Optimization Algorithms Considered for DRLE

The fact that the merit function $\phi(\theta)$ is inherently multimodal (*i.e.*, nonlinear and nonconvex) makes it difficult to obtain the global minimizer $\hat{\theta}$ in Eq. (8). Since no closed-form solution exists in this case, the simplest way to solve this problem is to use a linear LS (LLS) method which provides a nice closed-form solution. However, the solution is approximate since the nonlinear vector function $\mathbf{L}(\theta)$ must be linearized at some point $\hat{\theta}$ such that

$$\mathbf{L}(\theta) \approx \mathbf{L}(\hat{\theta}) + \mathbf{J}(\hat{\theta})(\theta - \hat{\theta}).$$

Then, we can solve the normal equations $\mathbf{J}^T \mathbf{J} \hat{\mathbf{h}} = \mathbf{J}^T \mathbf{r}$, where $\hat{\mathbf{h}} = \theta - \hat{\theta}$, to obtain the LLS solution [12]

$$\bar{\theta} = \hat{\theta} + \left(\mathbf{J}(\hat{\theta})^T \mathbf{J}(\hat{\theta}) \right)^{-1} \mathbf{J}(\hat{\theta})^T \mathbf{r}(\theta) \Big|_{\theta=\hat{\theta}}.$$

²This framework can readily be extended to a more general problem that includes the path loss gradient α optimized simultaneously with the source location by setting $\theta = [x \ y \ \alpha]^T$, leading to an $M \times 3$ Jacobian matrix $\mathbf{J}(\theta)$. We have examined this case and found the results to be quite acceptable.

If the evaluated point $\hat{\theta}$ is far from the global minimum, the first-order approximation does not accurately represent the actual function and no further improvement can be made. Due to this inefficiency and inaccuracy, an iterative method often needs to be employed to find the global minimizer.

There have been many iterative algorithms proposed specifically for nonlinear LS (NLS) optimization, exploiting the special structure of ϕ to enhance efficiency. We tested various methods including steepest descent, Gauss-Newton, Levenberg-Marquardt (LM) and trust region (TR), and found that the method of trust region using a subspace technique and preconditioned conjugate gradients [10], [13] performs best in terms of localization accuracy and robustness to a bad initial solution. It effectively tackles scenarios where LM may not work properly such as negative curvature and poor scaling. This approach finds the local minimizer in the constrained “trust region” whose size at the k th iteration Δ_k is adjusted according to its performance during the previous iteration. The DRLE-TR quadratic subproblem is

$$\theta_{k+1} = \min_{\theta_{k+1} \leftarrow h_k} \left\{ \mathbf{r}(\theta_k)^T \mathbf{J}(\theta_k) \mathbf{h}_k + \frac{1}{2} \mathbf{h}_k^T \mathbf{J}(\theta_k)^T \mathbf{J}(\theta_k) \mathbf{h}_k : \|\mathbf{D}_k \mathbf{h}_k\| \leq \Delta_k, \mathbf{h}_k \in \text{span}[\mathbf{J}^T \mathbf{r}, (\mathbf{J}^T \mathbf{J} + \gamma \mathbf{I})^{-1} \mathbf{J}^T \mathbf{r}] \right\},$$

where γ is appropriately chosen to ensure that $\mathbf{J}^T \mathbf{J} + \gamma \mathbf{I}$ is positive definite, and $\mathbf{D}_k$ is the diagonal scaling matrix [13]. All the results presented next are obtained by this approach.

V. ANALYSIS

This section is devoted to an analysis of DRLE and a comparison between DRLE and its RSS-based counterpart, RLE, with different degrees of correlated shadowing. Also, practical advantages of DRLE are discussed.

A. Numerical Analysis

Since a single start for the optimization can bias the comparison due to the nonconvex NLS error function of DRLE and RLE, we choose a multi-start option which picks the minimizer based on the outcomes produced from multiple starting points. Specifically, in this work, 10 random points are selected to assure the global convergence. The effect of node geometry on localization (*a.k.a.*, geometric dilution of precision [2]) is taken into account by randomly selecting $m+1$ anchor and source locations in all simulations. The location accuracy is shown in terms of the root mean squared error (RMSE) defined by $\sqrt{E(\|\theta - \hat{\theta}\|_2^2)}$ (meters) in a $30m \times 30m$ area.

In Fig. 2, the localization performance of DRLE and RLE with respect to the number of anchor nodes is presented for $\sigma_S = 5$ dB and $\alpha = 3$. The results are parameterized by the shadowing correlation to demonstrate its impact. In this analysis, the same correlation value is assumed for all the links in the network. Please note that an ideal assumption was made for RLE in that perfect information about the radio propagation parameters as in Eq. (1) was available (*i.e.*, $P_r(d_0)$ was known perfectly). We summarize our key findings as follows. First, higher correlation actually improves the location accuracy of

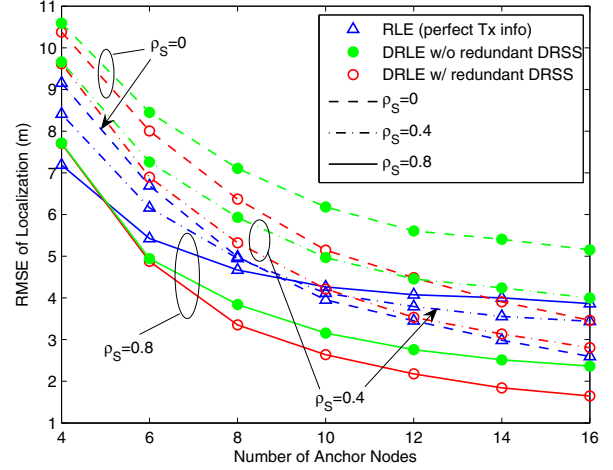


Fig. 2: RMS localization errors of RLE and DRLE versus the number of anchor nodes m ($\sigma_S = 5$ dB and $\alpha = 3$).

RLE when the number of anchors is small ($m < 9$), but deteriorates its performance when the number is increased beyond $m = 9$. The former conflicts with the previously known results (*e.g.*, [14]) based on the Cramer-Rao Lower Bound (CRLB) which assumes an unbiased estimator [12]. This is because when the number of anchors (*i.e.*, observations) involved is small, the estimates produced from the LS optimization for RLE are actually biased, and the correlation helps location estimation. On the other hand, as more anchors are added, RLE becomes unbiased and achieves the CRLB asymptotically so that the location accuracy becomes worse for higher correlation values. The second key observation from Fig. 2 is that the localization performance of DRLE is always better for higher correlations as expected from our previous discussion. When the correlation is high, in general, DRLE outperforms RLE. The third observation is that incorporating the redundant DRSS measurements is beneficial for localization. In particular, a greater advantage can be found with more anchors. Further, the benefit of an additional anchor node is higher as compared to RLE, thus making DRLE superior to RLE even with relatively low correlation. This is due to the fact that as more anchors are involved, the larger the number of redundant DRSS measurements that are available. As a result, its susceptibility to measurement and fading noise can be mitigated, as similarly found for TDOA [15].

In Fig. 3, the impact of log-normal shadowing variance on RLE and DRLE is shown by varying σ_S . Due to the two-sided localization behavior of RLE observed above, two cases of $m = 6$ and $m = 14$ are considered as shown in Figs. 3a and 3b, respectively. From this study, similar observations can be made. Specifically, for $m = 6$, correlation improves the localization performance of both RLE and DRLE, but the improvement rate of DRLE is higher. When the correlation is low ($\rho_S < 0.4$), RLE outperforms DRLE, whereas higher correlation makes DRLE superior to RLE. On the other hand, for $m = 14$, the correlation worsens the performance of RLE, whereas the localization accuracy of DRLE is considerably

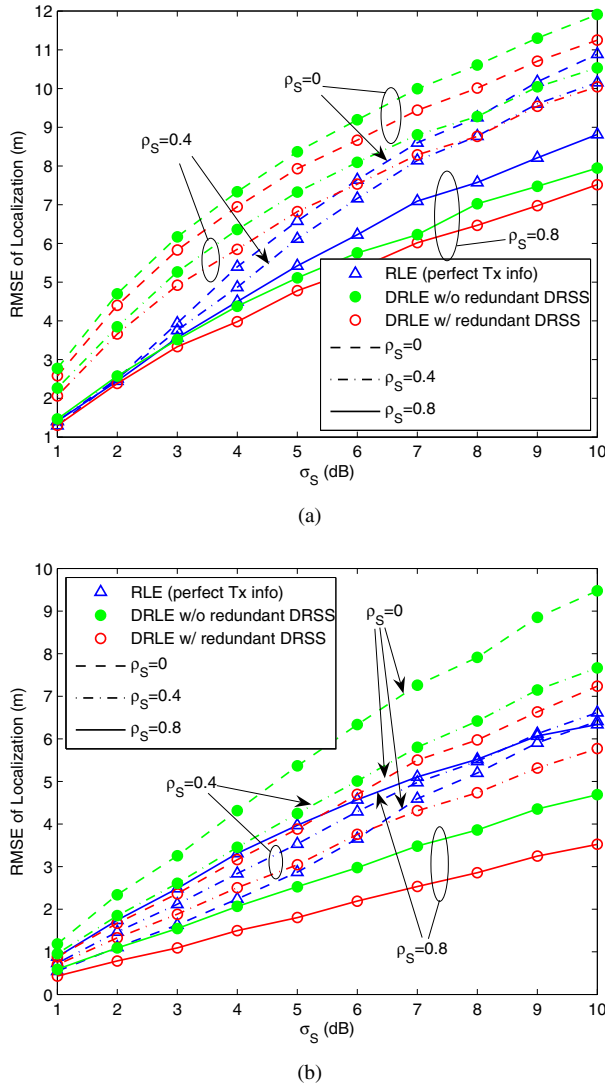


Fig. 3: RMS localization errors of RLE and DRLE versus Std. of shadow fading σ_S ($\alpha = 3$) for (a) $m = 6$ and (b) $m = 14$.

improved. We also see that the redundant DRSS information improves the performance of DRLE. Although the results with different values of path loss gradient α are not included here due to the paper size (the performance of both DRLE and RLE improves with higher values of α), similar observations regarding the correlated effects were found.

B. Practical Advantages of DRLE

The use of DRSS for localization is promising since it can significantly alleviate the passive dependence of localization on the source which could be defective/malicious or uncooperative (e.g., legacy users in cognitive radio networks), while retaining the advantages of RSS-based positioning. No or minimum control message overhead (e.g., periodic beacon signals transmitted at high power) between the source and neighboring nodes is required for localization. This not only saves bandwidth and energy but also conceals the localization process from the source which brings great benefits for

surveillance applications. It can also considerably reduce the effort of manual parameter calibration as well as transceiver complexity. Further, the shadowing environment with higher spatial correlation can improve the location accuracy of DRSS-based methods unlike many other positioning techniques whose performance would degrade.

VI. CONCLUSION

The problem of estimating the position of an unknown source, given a set of DRSS measurements from local anchors (i.e., DRLE), was presented from a geometric perspective and solved using an LS approach, making full use of the available DRSS information. This study showed that the location accuracy of DRLE improves as higher spatial correlation of shadow fading is encountered, where most existing positioning techniques may struggle. Further, it was found that while the spatial correlation deteriorates the localization performance of an RSS-based LS estimator when the number of anchor nodes is large (e.g., $m > 9$, if $\alpha = 3$, $\sigma_S = 5$ dB), its performance improves with smaller numbers of anchors.

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